

# Md. Tanvir Ahmed Tonmoy

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## Professional Summary

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Electrical and Electronic Engineering undergraduate with hands-on experience in RTL design, VHDL/Verilog, FPGA-oriented digital design, processor architecture, finite-state machines, testbench development, waveform-based verification, and CMOS/VLSI design. Currently designing a simplified RISC-V processor in synthesizable VHDL and an SRAM-based in-memory computing architecture in Xilinx ISE. Experienced with Cadence Virtuoso, Genus, Innovus, Python, and embedded hardware development.

## Education

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### BRAC University, Dhaka, Bangladesh

Bachelor of Science in Electrical and Electronic Engineering

Specialization: Electronics

CGPA: **3.72/4.00**

## Technical Skills

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### HDL / RTL

VHDL, Verilog, RTL Design, FSM Design, Synthesizable Design, Testbench Development, Functional Simulation

### Digital / VLSI

RISC-V Processor Design, SRAM Architecture, Compute-in-Memory

### Programming

Python, C, C++, MATLAB

### FPGA / EDA

Xilinx ISE, Cadence Virtuoso, Genus, Innovus, Maestro, Waveform Analysis, Timing Verification

### Embedded Systems

ESP32, STM32, Arduino, Teensy, Communication protocols, Wireless Modules

### Tools

GitHub, VS Code, LaTeX, Altium Designer, Autodesk Eagle, Proteus, PSPICE

## Relevant RTL, FPGA & VLSI Projects

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### SRAM-Based In-Memory Computing Architecture in VHDL

2026

- Designed and simulated an SRAM-based in-memory computing architecture using VHDL in Xilinx ISE.
- Developed memory, computational, and control logic with an FSM to coordinate memory and in-memory compute operations.
- Verified functionality using VHDL testbenches, simulation waveforms, and synthesized technology schematics.

### Variation-Aware SRAM Compute-in-Memory Architecture

2026

- Designed and simulated a 45 nm 6T SRAM-based compute-in-memory Boolean architecture in Cadence Virtuoso.
- Implemented AND, OR, NAND, NOR, XOR, and XNOR operations and evaluated transient behavior.
- Performed Monte Carlo timing analysis to quantify timing-induced logic errors under process variation.
- Developed a Python/PyTorch hardware-aware BNN evaluation flow linking circuit-level error probability with inference accuracy.

### 8T SRAM Crossbar and SRAM-Based Spiking Neural Network

2026 – Present

- Designing and characterizing an 8T SRAM cell in 45 nm CMOS and developing a 4×4 SRAM crossbar.
- Extending the architecture toward an SRAM-based spiking neural network for neuromorphic computing applications.

### VHDL–FPGA Based RISC-V Processor Design

2026

- Designing and implementing a simplified RISC-V processor using synthesizable VHDL for FPGA implementation.
- Developing modular processor components and integrating them into a complete processor architecture.
- Building a comprehensive VHDL testbench to verify supported instruction execution through simulation and waveform analysis.
- Preparing the design for synthesis, functional demonstration, and FPGA-oriented implementation.

### 45 nm CMOS Phase-Locked Loop

2026

- Designed a CMOS PLL in Cadence Virtuoso including PFD, charge pump, loop filter, VCO, and divide-by-48 frequency divider.

- Performed DC/transient analysis and evaluated lock behavior, frequency response, and current consumption.

## Engineering Experience

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### Industrial Attachment – Neural Semiconductor Ltd.

2026

Exposure to semiconductor design workflows including digital design, physical design, analog design, RF design, and fabrication-related processes.

### Electronics Subsystem Lead – BRACU Diganta

2024 – Present

Led onboard electronics design, sensor integration, telemetry hardware, power planning, and subsystem integration for model-satellite systems. Worked on CanSat 2025 hardware development; the team was selected as a global finalist in Virginia, USA.

### Undergraduate Teaching Assistant – BRAC University

February 2025 - June 2025

Supported undergraduate Electrical and Electronic Engineering coursework and student learning activities.

## Selected Publications

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- H. H. Jui, **M. T. A. Tonmoy**, et al., “Variation-Aware SRAM Compute-in-Memory Boolean Architecture for Binary Neural Networks,” *IEEE Computer Society Annual Symposium on VLSI (ISVLSI)*, 2026, Kolkata, India.
- F. Uddin, N. Shahrair, M. M. Rahman, **M. T. A. Tonmoy**, et al., “Soft Robotic Bellow-Controlled Arm Using Multi-DOF Control System,” *INTCEC 2025*, Chicago, USA.
- M. M. Chowdhury, M. M. Rahman, M. Mahmud, K. A. Rahim, **M. T. A. Tonmoy**, et al., “BRACU Diganta: An Affordable CanSat Learning Kit for Students in Developing Countries,” *International Astronautical Congress (IAC/IAF)*, 2025, Sydney, Australia.

## Selected Achievements

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- Dean’s List for six semesters.
- Merit-based tuition scholarship: 25% for four semesters and 10% for three semesters.
- CanSat Competition 2025 Global Finalist, American Astronautical Society, Virginia, USA.

## References

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